

[DC] Adaptive Visual Complexity in VR Training: An Eye-Tracking Approach to Detecting and Regulating Embodied Cognitive Load

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ABSTRACT

Virtual Reality (VR) provides an effective medium for training complex spatiotemporal procedures. However, current systems rarely account for individual differences in cognitive load during embodied interactions. This doctoral research investigates how integrated eye-tracking in VR headsets can be used both to detect and to regulate learners' embodied cognitive states. First, I will develop an instrument that combines low-level ocular metrics (e.g., pupil dilation, fixation) with higher-level gaze behaviors (e.g., gaze entropy, scanpath organization) to model embodied cognitive load in spatiotemporal VR tasks. Using these models, I will design and evaluate a visually adaptive VR training environment that adjusts visual complexity, saliency, and scene structure in real time based on inferred user state. Two planned user studies will (1) validate the embodied cognition instrument and train real-time classifiers, and (2) test whether state-contingent visual adaptation improves learning and user experience. This work aims to contribute eye-tracking-based models and design guidelines for adaptive VR training.

Index Terms: Adaptive, virtual reality, eye tracking, machine learning

1 INTRODUCTION

Virtual Reality (VR) has become a powerful medium for training complex spatiotemporal procedures, such as assembly, disassembly, and navigation, offering high ecological validity, repeatability, and fine-grained control over task difficulty and feedback [10]. However, learners in VR experience substantial variability in Cognitive Load (CL). Cognitive Load Theory (CLT) argues that learning is maximized when task demands stay within a bounded region of working-memory load [9], echoing the Yerkes–Dodson law that performance deteriorates when arousal is too low (boredom) or too high (overload) [12]. Yet most current VR training systems remain static: they do not sense the learner's moment-to-moment state nor adapt the environment accordingly.

Subjective instruments such as NASA-TLX are widely used to assess CL, but are retrospective, interruptive, and rely on learners' self-awareness and recall [10]. They cannot support fine-grained, real-time adaptation. Physiological and behavioral signals offer a complementary path: prior work has shown that pupil dilation, fixation duration, and related eye-tracking features can infer CL in VR [10, 5]. Recent studies have demonstrated that models trained on such markers can predict CL continuously during immersive tasks [10]. Integrated eye-tracking is becoming standard in head-mounted displays, lowering the barrier to deploying signal-based adaptation in real-world VR systems [2].

Yet two important gaps remain. First, existing eye-tracking-based CL instruments rarely capture embodied CL, despite their centrality in spatiotemporal VR tasks. Most prior work uses abstract tasks (e.g., n-back) or measures CL in static contexts where

users passively view stimuli. In contrast, embodied cognition theories emphasize that cognition is situated and tightly coupled to continuous bodily action, and the environment is part of the cognitive system [11]. In VR, learners must simultaneously coordinate head movements for scene exploration, hand movements for object manipulation, and gaze for visual search—all while navigating cluttered 3D spaces. These embodied demands create unique patterns of attentional allocation and motor-cognitive coordination that are not captured by traditional CL tasks. For example, a learner may exhibit high pupil dilation not due to memory load, but rather from coordinating visual search with reaching and grasping in a cluttered workspace. Existing instruments do not systematically manipulate or measure these embodied interaction demands.

Second, visual complexity, including object density, distractors, contrast, and background structure, is a primary driver of CL in spatiotemporal VR tasks, yet most adaptive systems do not directly manipulate it [3, 7, 1]. Most adaptive VR systems instead focus on high-level instructional content (e.g., LLM-generated hints) or coarse task difficulty changes [8]. A learner overwhelmed by visual clutter may struggle to locate objects regardless of the clarity of instructions. Visual adaptation offers complementary advantages: it is immediate, language-independent, directly traceable in eye-tracking data, and can be precisely controlled to regulate CL in real time. To address these gaps, my research is centered around answering these questions:

- **RQ1 (Signal Level):** Which low-level eye-tracking features (e.g., pupil dilation, fixation duration, saccade metrics) enable robust real-time classification of embodied CL states (low/high) in VR training environments?
- **RQ2 (Behavioral Level):** How do higher-level gaze behaviors (e.g., gaze entropy, scanpath organization) distinguish between low and high embodied CL states during VR training?
- **RQ3 (User Modeling):** How can combined signal-level and behavioral gaze metrics be integrated to model users' embodied cognitive states that generalize across tasks and individuals?
- **RQ4 (Adaptation):** Which visual adaptation strategies—reducing complexity under high CL and increasing complexity under low CL—most effectively improve learning outcomes and user experience in VR training?

2 RESEARCH PLAN

My dissertation is structured in two major thrusts: (1) modeling embodied CL from eye-tracking signals in spatiotemporal VR training, and (2) designing and evaluating visually adaptive VR environments that use these models to adjust visual complexity and regulate low/high embodied CL.

2.1 Modeling Embodied Cognitive Load

Signal-Level and Behavioral Level: Building on prior work using pupil dilation and fixation duration to predict CL in VR [5, 6, 4],

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I will extend signal-level modeling in two ways. First, I will compare low-level eye-tracking features (pupil dilation, fixation duration, saccade amplitude, microsaccade rate) for real-time classification of low and high CL. Second, I will incorporate behavior-level gaze features such as gaze entropy, scanpath compactness, and spatial distribution of attention. I will use supervised machine learning trained on eye-tracking features, task performance, and subjective CL ratings. The goal is to develop models that are accurate, robust, and lightweight enough to run on commodity VR hardware.

Embodied Cognitive Load Instrument: I will develop an *Embodied Cognitive Load Instrument* tailored to spatiotemporal VR tasks. This instrument will consist of short VR scenarios where participants localize, track, and manipulate objects in a 3D workspace. I will manipulate: (1) visual complexity (object density, spatial arrangement, background clutter, contrast) and (2) embodied interaction (coordination required between head, hand, and gaze). Participants will complete all conditions in a within-subject design using an eye-tracking-enabled HMD. Dependent measures include eye-tracking features, task performance (completion time, errors), and subjective measures (NASA-TLX, presence). I will use factor analysis to derive latent dimensions that capture embodied cognitive demands beyond traditional CL constructs. This instrument serves as both a calibration tool for adaptive systems and a reusable research asset.

2.2 Visually Adaptive VR Training Environment

The second thrust focuses on designing and evaluating an adaptive VR training environment for complex spatiotemporal tasks (e.g., advanced manufacturing assembly). Here, the embodied CL instrument and learned models serve as the backbone for a *user CL estimator*. The adaptive system will continuously monitor eye-tracking and performance data, infer the learner's current CL (low, high), and adjust visual design parameters accordingly.

2.3 User Studies

I plan to conduct two user studies:

Study 1: Calibration and Validation of the Embodied Cognitive Load Instrument. Participants will perform the instrument tasks in a controlled VR environment. Goals are to (1) validate that visual complexity manipulations affect CL indicators; (2) identify eye-tracking features that reliably discriminate low and high CL; and (3) train and evaluate real-time classification models.

Study 2: Evaluation of Visually Adaptive VR Training. This study will compare three conditions: adaptive (visual complexity adjusts based on inferred CL), non-adaptive (static complexity), and reverse-adaptive (complexity increases under high CL, decreases under low CL). Outcome measures include learning gains, objective and subjective CL, and user experience. This study tests whether state-contingent visual adaptation keeps learners near optimal CL and improves performance and experience.

3 CURRENT STATUS AND EXPECTED CONTRIBUTIONS

To date, I have completed two user studies: one developing machine learning models (MLP and RF) using pupil dilation to detect CL in VR manufacturing training, achieving 0.84 and 0.72 accuracy for binary CL classification [5]. Another is examining gaze behavior—particularly gaze entropy—to understand users' strategies in advanced manufacturing and visually complex learning environments. Preliminary results indicate that eye-tracking features and gaze entropy can predict subjective CL and differentiate effective from ineffective problem-solving strategies.

The expected contributions of this work are: (1) a validated VR-based *Embodied Cognitive Load Instrument* that detects embodied CL for spatiotemporal tasks using eye-tracking, performance, and subjective measures; (2) real-time models that combine signal-level and behavior-level gaze features to infer CL; (3) a visually

adaptive VR training framework demonstrating how manipulating visual complexity and saliency based on inferred state can keep learners at an optimal CL level; and (4) design guidelines for gaze-informed, visually adaptive VR training systems, complementing existing work on multimodal XR fidelity and adaptive interfaces.

4 CONCLUSION

This doctoral research aims to advance adaptive VR training by connecting embodied CL, gaze-based user modeling, and visual design. By developing an embodied CL instrument and a visually adaptive VR environment, the project seeks to show how integrated eye-tracking can be used not only to *detect* CL but also to *regulate* it through real-time visual adaptation. Participation in the Doctoral Consortium will provide critical feedback on (1) the ecological validity of the embodied CL instrument, (2) the design of visual adaptation that balance CL and learning engagement, and (3) strategies for deploying lightweight classification models on resource-constrained VR hardware in real-world training contexts.

REFERENCES

- [1] F. Chiossi, Y. Turgut, R. Welsch, and S. Mayer. Adapting visual complexity based on electrodermal activity improves working memory performance in virtual reality. *Proceedings of the ACM on Human-Computer Interaction*, 7(MHCI):1–26, 2023. 1
- [2] M.-P. Funk, N. Navab, and L. Schütz. Eye tracking-based adaptive user interfaces in virtual reality eye surgery training. In *Proceedings of the 2025 Mensch und Computer 2025*, pp. 578–582. 2025. 1
- [3] I. Kartiko, M. Kavakli, and K. Cheng. Learning science in a virtual reality application: The impacts of animated-virtual actors' visual complexity. *Computers & Education*, 55(2):881–891, 2010. 1
- [4] M. Nasri. Towards intelligent vr training: A physiological adaptation framework for cognitive load and stress detection. In *Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization*, pp. 419–423, 2025. 1
- [5] M. Nasri, M. Kosa, L. Chukoskie, M. Moghaddam, and C. Harteveld. Exploring eye tracking to detect cognitive load in complex virtual reality training. In *2024 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 51–54. IEEE, 2024. 1, 2
- [6] M. Nasri, U. Narayan, M. F. Sonbudak, A. Simonson, M. Chiu, J. Donati, M. Sivak, M. Kosa, and C. Harteveld. Designing a virtual reality training apprenticeship for cold spray advanced manufacturing. In *2024 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 541–544. IEEE, 2024. 1
- [7] M. Rettinger, X. Jiang, J. Yang, and G. Rigoll. Visual complexity in vr: implications for cognitive load. In *2024 IEEE conference on virtual reality and 3d user interfaces abstracts and workshops (VRW)*, pp. 783–784. IEEE, 2024. 1
- [8] J. Santhosh, A. Dengel, and S. Ishimaru. Gaze-driven adaptive learning system with chatgpt-generated summaries. *IEEE Access*, 12:173714–173733, 2024. 1
- [9] J. Sweller. Cognitive load during problem solving: Effects on learning. *Cognitive science*, 12(2):257–285, 1988. 1
- [10] J. Wei, E. Siegel, P. Sundaramoorthy, A. Gomes, S. Zhang, M. Vankipuram, K. Smathers, S. Ghosh, H. Horii, J. Bailenson, et al. Cognitive load inference using physiological markers in virtual reality. In *2025 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pp. 759–769. IEEE, 2025. 1
- [11] M. Wilson. Six views of embodied cognition. *Psychonomic bulletin & review*, 9(4):625–636, 2002. 1
- [12] R. M. Yerkes, J. D. Dodson, et al. The relation of strength of stimulus to rapidity of habit-formation. 1908. 1